



Full-Stack Software Engineer - AI Systems

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Professional Profile

Full-stack software engineer building controlled AI inside real products: authenticated Gemini function tools, document-grounded retrieval, usage and cost accounting, retention-aware chat, and OR-Tools constraint optimization. Connects React and Next.js interfaces to PHP, Node.js, Python, relational data, and production security boundaries.

Education

BSc (Hons) in Software Engineering
University of Plymouth, UK
2022 - 2025

IT Foundation Programme
NSBM Green University, Sri Lanka
2020 - 2021

Skills

AI & Optimization

- Gemini function calling
- RAG / File Search
- OR-Tools CP-SAT
- Tool policy controls
- Usage / cost accounting
- Retention controls
- Local speech design

Data & Security

- MySQL / PostgreSQL
- Entra ID / Auth.js
- RBAC
- Rate limiting
- Sanitization / HMAC
- Origin controls
- Audit state

Full Stack

- TypeScript / Node.js
- Next.js / React
- PHP
- Python / FastAPI
- REST APIs
- Flutter
- Angular

Operations

- Docker / Compose
- Usage accounting
- Migrations
- CI/CD
- Apache / PM2
- Testing
- Observability

Experience

NSBM University - Associate Software Engineer 2025 - Present

- Build AI-enabled student and staff workflows as part of the NSBM development team, from retrieval and function tools to operator controls.
- Design service and data boundaries for identity, authorization, read-only integrations, usage accounting, retention, and deployment.

Iplay Global - UI Developer Intern 2022 - 2023

- Implemented responsive interfaces and integrated frontend workflows with REST APIs.

Infigo Developers - Developer 2020 - 2022

- Delivered client websites and developed practical React and Angular foundations in a startup team.

Highlighted Projects

GreenMate AI - Next.js, Gemini, Auth.js, MySQL

- Implemented authenticated function tools for student data with server-owned identity and two-stage policy enforcement.
- Persisted model/token usage and cost signals, with sanitization, rate limiting, session ownership, and retention-aware history.

NSBM RAG Bot - PHP, Gemini File Search, MySQL

- Built document-grounded retrieval, an embeddable desktop/mobile widget, and administrative dataset/session tooling.
- Added HMAC request signing, origin allowlisting, sanitization, database rate limits, and usage reporting.

Timetable Generator Gen2 - OR-Tools, FastAPI, Next.js

- Encoded documented academic, lecturer, room, and time constraints in a full-semester CP-SAT model.
- Isolated compute behind a private service boundary with read-only UMIS access and explicit allocation conflicts.

Doc-Assist Pro - Node, React Native, Angular

- Built as a team across patient, doctor, administrator, appointment, and medical-record workflows.
- Implemented a doctor-recommendation workflow over the shared Node and PostgreSQL backend.